Trustworthy AI for Citizen-Centric Digital Governance and Intelligent Public Services: Lessons from a Societal Innovation Collaboration Portal for Jharkhand

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Abstract

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Citizens' requests are being digitized all over the world, and Artificial Intelligence (AI) is increasingly used to classify, prioritize and route citizens' requests. But, not only their technology, but their trustworthiness, gives them legitimacy. This chapter establishes a framework on trusted AI, using the pillars of transparency, fairness, accountability and explainability, which are interdependent. The state of Jharkhand, India, has a proposed case of a digital portal called Societal Innovation Collaboration Portal, where citizen reported societal problems are classified and routed to the university, industry and government departments where the problems are evaluated and solved by an AI system. The chapter emphasizes design approaches that can be employed to make the system trusted but not simply assumed trusted, such as: human review with confidence thresholds, district accountability with auditability, milestone transparency, and fairness monitoring. Additional potential risks to trust in such systems are discussed and a risk matrix that can be used to mitigate risk is provided. The chapter describes trustworthy AI as a cycle of consent, explainable automation, institutional responsibility and audibility of the public, rather than a compliance checklist.

Keywords

Explainable AI, Algorithmic Accountability, NLP, Multi-stakeholder Innovation Platforms, NEP 2020, Smart Governance, Citizen-centred Governance, Trust-worthy AI, e-governance.

Introduction

In the past few years, public administration systems have been shifting from e-governance systems with static resources and services to systems that rely on algorithms to determine the classification of complaints, prioritize responses to complaints and even suggest what action the institution should take based on the complaints [1]. A sense of efficiency is plain and there are clearly benefits to doing this, in the past requiring multiple clerical staff to manually wade through citizens' complaints, now, with NLP and machine learning models can be accomplished on a large scale. There's more to it than efficiency, though. The emerging literature on algorithmic government emphasizes the risks that come along with the automation of public services: in particular, the risk of opacity, bias, and poor accountability of decision-making; people who are unaware of how the process works and mistrust the way decisions

were made are less likely to participate in the process and reveal their true intentions [1], [8]. Not secondary but primary like many in rural and semi-urban parts of India, people have different levels of digital literacy, different linguistic backgrounds and have lost trust in the bureaucracy procedures as well.

This chapter reads the following: that trustworthy AI is not an additional ethics requirement, to the citizen-centred approach to digital governance, but it is a key requirement if the paradigm is to be used successfully. To capture the principles of trustworthy AI design, which have been defined in the international standards of trustworthy AI design [2, 8], the chapter suggests a governance framework specifically geared towards the digital platforms used for interactions between citizens, government bodies, academia, and industry. For this, a concrete yet emerging case study, Societal Innovation Collaboration Portal (SICP) and its motivation to crowdsource societal issues being faced by citizens and channelize their solutions to academic and industrial partners via AI based problem categorization [10] is discussed.

The remainder of the chapter is structured in the same way. In section 1, the conceptual understanding of the concept of trustworthy AI as it relates to citizen-centric digital governance is introduced and the proposed approach to developing an AI trustworthy framework is distinguished from the existing approach of e-governance that is based on data. Section 2 discusses the issue of engineering trust within multi-stakeholder platforms' architecture using Societal Innovation Collaboration Portal as an example. In section 3, the strategic pillars that are being used in the life cycle of the platform are described: transparency, accountability and fairness. In Section 4, the risks and ethical obligations that any such system will have to face are critically examined, such as the reliability of the models, data privacy and the digital divide. Finally, the chapter advocates for considering the trustworthy AI in public-service platforms as a governance loop to be reinforced on an ongoing basis.

1. Foundations of Trustworthy AI in Citizen-Centric Governance

Understanding of the nature and the need of trustworthy AI in public services delivery goes beyond the basics; it calls for a more fundamental understanding of what it means and why the current digital governance paradigms have not been able to attract it. It discusses the potential of e-governance for service delivery via AI, the principles of trustworthy AI, and analyzes the issue of loss of trust that has constrained citizen access to AI systems for addressing grievances and services.

1.1 From e-Governance to AI-Augmented Public Service Delivery

Most of the initial e-governance initiatives have been about converting the current bureaucracy into electronic forms, replacing existing paper forms with web forms, replacing existing paper registers with databases, and allowing citizens to file their requests online rather than by mail. The decision logic, however, did not change, it was mostly the same and human. The second type of AI-augmented governance is a qualitatively higher one where software does not only

record a citizen's request, but it also interprets it, categorises and routes the request; formerly, human officials did that by hand. This sort of algorithmic government can have clear efficiency gains, as automated classification can be done in a more regular way, and in a more consistent way, thousands of times more than human experts would be able to do classification in a time-consuming and repetitive way, and instead make more nuanced and complicated judgments. Yet, the very same metamorphosis introduces a significant amount of interpretive power in software that is often not comprehensible to the citizens for which it exists, and where the very state of trust is most fragile.

1.2 Core Principles of Trustworthy AI: Transparency, Fairness, Accountability, Explainability

International guidance on trustworthy AI, such as the European Commission's Ethics Guidelines for Trustworthy AI and the NIST AI Risk Management Framework, coincide in a limited number of interrelated principles which are equally relevant to applications to citizen-facing public service systems [2,6]. Transparency demands clarity and enables citizens and oversight bodies to grasp, to a reasonable degree of detail, how and why a system ended up within a certain classification or routing. The basic principles of fairness is that the system does not systematically disadvantage specific districts, dialects, socio-economic and/or cultural groups. Accountability demands that the responsibility for an automated decision must always be attributed to a specific human process or institution, and never be an accountability void. Explainability, which is closely related to transparency, means that the internal logic of a model, no matter how complex, can be expressed with ease in terms that a non-technical citizen or auditor can easily follow [8]. More importantly, these principles are not mutually disjointed items that a system could choose to do or not; a system can be transparent in a biased process, or explainable in an unaccountable process. All four principles are needed to work together to create a governance framework for trustworthy AI, which is properly understood.

1.3 The Trust Deficit in Conventional Grievance and Service-Delivery Systems

Regardless of the fact that they have been digitized, conventional public grievance systems have failed to establish a long-term citizen trust for certain technical and non-technical reasons. Many citizens report that they have made complaints in systems which do not show them any feedback loop, without knowing if their complaint was received, understood, or acted upon [4]. Researchers working in the field of algorithmic decision-making in public administration have also found that vulnerable and marginalized citizens are more likely to be impacted by hidden failures of automated decisions, as they may not have the social or technical capital to challenge a seemingly opaque decision [7]. In such a state where majority of citizen identified problems are coming from rural, tribal and semi-urban communities, where the direct access to formal institutions has been mediated by intermediaries, this lack of trust is further exacerbated. So, creating an AI-powered application in this context can't be an after-thought, as trust needs to be embedded within the architecture of the application right from the beginning, as will be explained in the next section.

2. Engineering Trust into a Citizen-Centric AI Platform: The Societal Innovation Collaboration Portal

Having clearly identified the need to not take trust for granted, the section looks closely at how it can be actively built into the design of a multi-stakeholder AI platform. As a design case, it utilises the Societal Innovation Collaboration Portal, a digital platform that is proposed for the Government of Jharkhand, to connect citizen-reported societal challenges to university and industry-led solutions, in order to make abstract trust principles concrete.

2.1 Platform Overview and a Trust-by-Design Architecture

The Societal Innovation Collaboration Portal is intended to enable the citizens, community organizations, Panchayati Raj Institutions, Urban Local Bodies, and government departments to submit the societal challenges (related to education, healthcare, agriculture, water management, sanitation, and public infrastructure) with photographs, videos, geolocation and supporting documents [10]. The platform's architecture takes a different approach from viewing AI as a bolt-on add-on to a current workflow, but rather sees an AI categorization and routing engine as one component in a longer, more human-supervised process that includes problem validation, institutional assignment, multidisciplinary team formation, industry collaboration, milestone tracking, and public reporting. For trust: The AI system never makes a final decision on a citizen's submission without a review. Rather, it generates a recommendation, which can be confirmed, denied, or pushed up the list of recommendations, by downstream human actors, such as moderators, university coordinators or government officials.

2.2 AI-Enabled Categorization and Routing as a Trust Mechanism, Not Just an Efficiency Tool

The platform's AI layer performs three linked functions: automatic thematic categorization of incoming challenges using natural language processing, duplicate detection through semantic similarity to consolidate repeated reports of the same underlying issue, and priority scoring that weighs severity, frequency, and community endorsement. Framed only as an efficiency mechanism, this layer risks being perceived as a filter that decides which citizen voices are heard. Framed instead as a trust mechanism, its design choices change: submissions below a defined classification-confidence threshold are automatically routed to human moderators rather than silently auto-categorized; duplicate-detection merges are logged and reversible rather than silently discarding a citizen's individual submission; and priority scores are computed from transparent, documented weightings rather than an opaque composite index. These are the same technical components found in many algorithmic government systems [1], but their trust value depends entirely on whether their thresholds, logs, and weightings are made visible to oversight actors and, where appropriate, to citizens themselves.

2.3 Human-in-the-Loop Oversight and Explainable Routing Decisions

A unique aspect of the proposed architecture is that university coordinators assigned a challenge can view not just the AI categorization output, but a short explanation (matched keywords, confidence score, and expertise tags that would make the routing recommendation). This human-in-the-loop design does not necessarily make automation redundant but rather is an instance in which automation is limited to recommendations that are subject to contestation by an accountable human reviewer, in line with the developing literature on explanation-based accountability in public sector AI, which emphasizes that explanations need to be used as governance tools, not technical tools [9]. Figure 1 shows this design as a continuous governance loop and not as a linear pipeline, highlighting that citizen trust is not built once at the time of submission but is built and lost at every step of how the platform is operated.

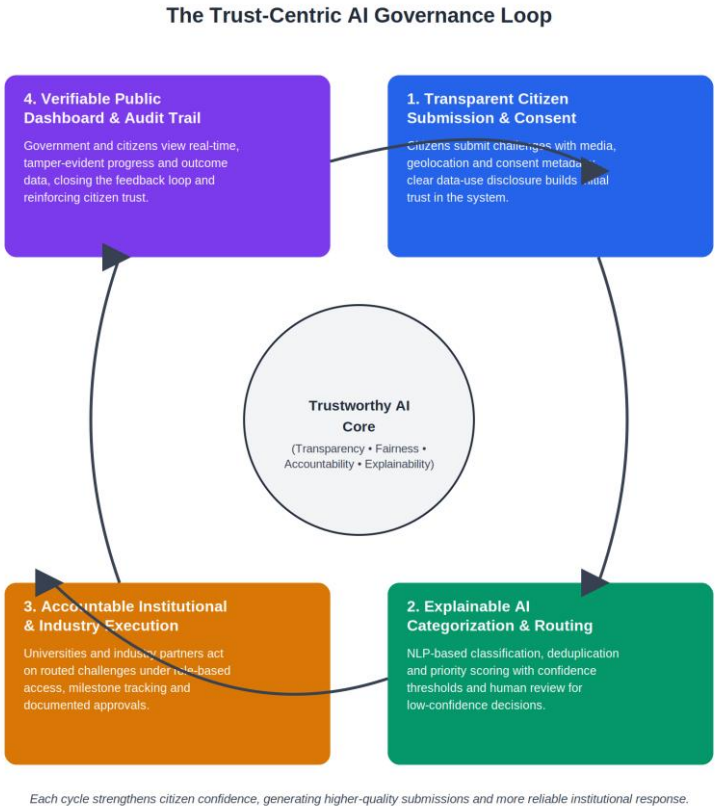


Figure 1: The Trust-Centric AI Governance Loop for the Societal Innovation Collaboration Portal

3. Strategic Pillars for Trustworthy Public-Service AI

This section builds on the architectural design discussed above, and identifies three strategic themes that need to be pursued throughout the platform's life (and not just at the initial design stage): Transparency, Accountability, Fairness. Specific mechanisms of the Societal Innovation Collaboration Portal are used to illustrate each pillar.

3.1 Transparency by Design across the Problem-to-Solution Lifecycle

Transparency goes beyond just providing a privacy policy or an AI-use disclosure. It demands that a citizen who raises a challenge may at any time be able to see the current status of that

challenge (received, categorized, routed, under university review, in active project development or resolved) without having to contact an official directly. Government departments, meanwhile, need transparency into aggregate patterns, such as the frequency of human overrides of AI routing recommendations and the number of unresolved challenges in individual domains and districts, which may indicate either that the model is under-trained or a need to retrain or recalibrate it. The ability to demonstrate this kind of responsiveness has been variously correlated with sustained citizen engagement with e-participation platforms and not just with the sophistication of technology [4] [5].

3.2 Accountability through Auditability, Role-Based Access, and Milestone Tracking

Accountability is achieved by role-based access control, where the university coordinators, industry partners and government administrators can only see and act on data pertinent to their role; and by the fact that every project that comes from a citizen-reported challenge has a milestone and approval path that can never be altered. If a challenge is merged as a duplicate, routed to a specific university, reduced to a lower priority, or otherwise, the action is recorded with the AI engine that triggered it (if it is under a documented threshold rule) or with a human moderator so that there is no action taken by the AI without a point of responsibility. The design addresses practitioners' concerns, highlighted in systematic reviews of AI accountability in the public sector, that accountability is hard to identify within automated systems because it is spread across many actors [14].

3.3 Fairness in Algorithmic Prioritization across Districts and Domains

Given that the platform combines challenges from all the different districts, regions, and thematic areas of Jharkhand, it is structurally vulnerable to the fact that fewer, less detailed, or non-Hindi/English language reports are submitted within these communities, rather than the challenges being less serious; it is exposed to the risk of systematically under-serving communities that submit less than others, whose reports are less detailed, and whose reports are not in Hindi/English. The periodic fairness audit of outcome distributions across districts, languages and channels of submission is thus not a compliance chore, but part and parcel of the operation, in line with the general recommendations from the algorithmic-governance literature for conducting an evaluation of automated public systems for disparate impact instead of intent [7]. The risks to this fairness objective, along with the platform's other key risks to trust, are plotted on a probability-impact matrix in Figure 2, which can help identify the areas of platform governance where focus should be directed.

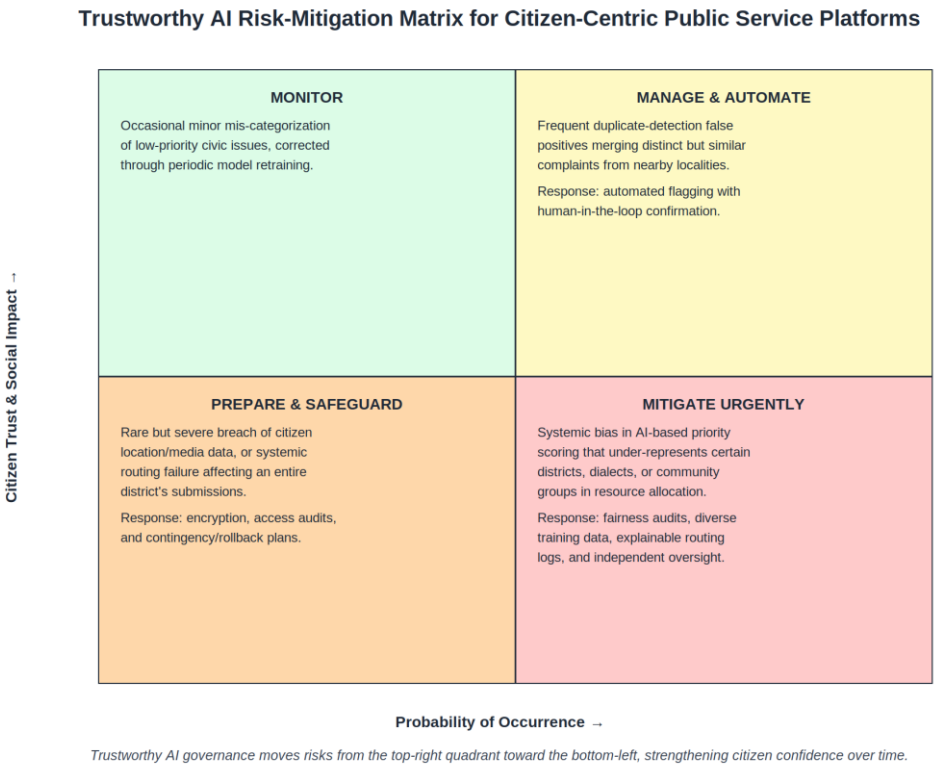


Figure 2: Trustworthy AI Risk-Mitigation Matrix for Citizen-Centric Public Service Platforms

4. Challenges and Ethical Imperatives in Deploying Trustworthy AI for Public Services

Even the best and most citizen-friendly AI platform comes with risk. In this section, three types of challenges are critically examined that the Societal Innovation Collaboration Portal, and platforms such as it, must actively address: the reliability of models and models' bias; data privacy in a multi-stakeholder data-sharing environment; and risk of exacerbating digital divides.

4.1 Managing Model Bias, Hallucination, and Reliability in Civic AI

When written feedback is ambiguous or code-mixed, these models may misclassify them; in addition, when they use generative components to summarize long citizen descriptions, they can sometimes generate a plausible-sounding but incorrect summary, a risk similar to the one of model hallucination in generative AI applications in general. In a civic sense, this type of mistake could send a true public-health threat to the wrong department or minimize its priority. Confidence-threshold escalation, retrieval-grounded summarization and required human verification of high-severity categories discussed in explainable AI literature are directly applicable and should be considered as safeguards and not enhancements [8].

4.2 Citizen Data Privacy and Consent in a Multi-Stakeholder Ecosystem

The platform has to balance transparency (wanting to share open data for institutional assessment) with privacy (wanting to minimize exposure of personally identifiable and

location-sensitive data) since data is shared across universities, industry partners, and government departments, including photographs, videos, and accurate geolocation of citizens. The issues of data ownership and consent in AI-enabled public systems also come to bear here: citizens need to be made aware of who can view their submission, and sensitive personal identifiers should be removed from the problem statement that is made public and shared with external university and industry reviewers where this is not needed for institutional evaluations [3].

4.3 Bridging the Digital Divide for Inclusive Trust

Without meaningful access by the population of interest, any platform that relies on citizen participation is not trustworthy. It is a documented fact that there are disparities in penetration of mobile phones, data connectivity, and digital literacy in rural and tribal areas of Jharkhand vis-à-vis urban areas. A trust-by-design approach must also therefore include supporting access mechanisms, such as facilitation with Common Service Centres, facilitators at the panchayat level and multilingual voice interfaces, so that the innovative use of AI does not further exacerbate existing inequities in public service access but, on the contrary, ensures that it is able to reconcile them.

Conclusion

The chapter has claimed that trustworthy AI is a basic requirement and not an added luxury of citizen-centric digital governance. With the proposed Societal Innovation Collaboration Portal for Jharkhand as a design case it has demonstrated how the principles of transparency, fairness, accountability, and explainability can be brought to life in architectural terms: The use of a confidence threshold to review AI categorisation by humans, the use of reversible duplicate detection and logging of all copies, the auditability of all routing and approval decisions by role, and the use of periodic fairness audits across districts and languages. As the chapter has demonstrated, these protections are not risk free, however—they must be continually assessed to a living risk-mitigation matrix, tracking model reliability, data privacy, and digital-access equity as the platform expands from a pilot to a state-wide rollout.

The overarching takeaway for citizen-focused public-service platforms is that trust can't be certified and then assumed. It's more commonly referred to as a governance loop: citizens submit their information in the transparent manner, and as the result of this, the initial confidence is established, which is then reinforced by the fact that the system processes information in explainable ways, the institutions and industries making decisions are accountable, and a verifiable public dashboard ensures that it can be confirmed that the system acted as it was supposed to do, by citizens and oversight bodies alike. As the public institutions, universities, and industry partners of this kind increasingly work together, this trust loop will become more and more embedded in the technical architecture more than just as a

communications exercise on top of the tech, creating a social impact that depends on sustained civic participation will require.

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